

C.A.R.E: Community Safety Incident Reporting System

A system for citizens to report incidents of violence and abuse, and for the corresponding authorities to be notified, dispatched, and held accountable. It was built for low-connectivity and bilingual deployment in Cameroon.

1 Introduction

This document defines the functions the system C.A.R.E (Community Alert & Reporting Engine) will be able to perform in its first version.

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for, a community safety incident reporting system. It is intended for the implementation of the system, and for any reviewer evaluating the design against the stated objectives.

1.2 Problem Statement

In Cameroon, citizens who witness or experience incidents such as violence, abuse, including child abuse, kidnapping, gender-based violence, and femicide, often lack a fast, reliable channel to alert the appropriate authority. Existing reporting relies heavily on direct phone calls or in-person visits, which assumes the reporter knows the right hotline for the type of incident, has network coverage to place a call, and is willing to be identified, which is a barrier for witnesses reporting sensitive cases like abuse or domestic violence.

On the response side, authorities have no unified system to receive, triage, and act on reports by type. Reports made by phone can be missed if a line is unattended, and there is no automatic escalation if a report goes unacknowledged, no photographic evidence attached, and no auditable record of what action was taken or when.

This creates a gap on both ends: citizens without a low-barrier, connectivity-tolerant way to report an incident with evidence, and authorities without a reliable, accountable system that guarantees a report reaches the correct responder quickly and escalates automatically if no one responds.

C.A.R.E addresses this gap by providing a structured incident-reporting system that routes each report to the correct authority group based on incident type, delivers redundant real-time alerts (in-app and email), and maintains an auditable status trail from submission to resolution, while keeping the final decision to act with human authorities.

1.3 Scope

C.A.R.E allows a citizen to report a safety incident with photo evidence and location through a mobile or web form. The report is routed automatically to the government or law-enforcement group responsible for that incident type, who are alerted in-app and by email, with an automatic SMS escalation if the report goes unacknowledged. Authorities update the incident's status as they act on it. The citizen can track that status without needing to be contacted directly.

C.A.R.E does not replace a country's emergency dispatch infrastructure. It is a reporting and alerting layer that sits in front of human decision-makers. No automated action is taken beyond notification.

1.4 Definitions & Abbreviations

TERM	MEANING
Reporter	A citizen submitting an incident report, authenticated or anonymous.
Admin	A member of an authority group (police, social welfare, fire service, etc.) who receives and acts on reports.
Routing group	The admin group an incident type is mapped to (e.g. child protection).
Escalation	The automatic re-alert and SMS fallback triggered when a report is not acknowledged in time.
JWT	JSON Web Token — the bearer credential issued by Cognito after sign-in.

1.5 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications
- AWS service documentation: Cognito, API Gateway, Lambda, DynamoDB, S3, SNS, SES, Secrets Manager, CDK

02 Overall Description

2.1 Product Perspective

C.A.R.E is a new, standalone system composed of a citizen-facing client, an admin dashboard, and a serverless backend on AWS. It is not integrated with any existing government case-management system in this version. Integration with any government case-management system will be considered in the next version.

2.2 Product Functions

Report intake

Structured incident submission with type, description, location, and photo evidence, witnesses (if any).

Role-based access

Separate citizen and admin identities, admins further grouped by authority type.

Live alerting

In-app + email notification the moment a report matching an admin's group is filed.

Escalation

Automatic SMS fallback when a report sits unacknowledged past a set threshold.

Status tracking

Auditable lifecycle from submission to resolution, visible to the reporter.

Secrets isolation

Third-party escalation credentials held outside application code.

2.3 Assumptions & Constraints

- The list of incident types is provisional pending frequency data specific to Cameroon and may be revised before implementation freezes.
- The client application must tolerate intermittent connectivity; report submission queues locally and syncs on reconnect.
- The interface is bilingual (French / English) as per Cameroon's official languages.
- SES operates in sandbox mode during development, so recipient addresses must be pre-verified until production access is granted.

03 User Classes & Roles

ROLE	IDENTITY POOL	CAN DO
Citizen	CitizenUserPool	Submit reports, upload evidence, view status of own reports (or an anonymous tracking ID).
Admin	AdminUserPool	View reports routed to their group, acknowledge, update status, add case notes.

Supervisor	AdminUserPool (elevated group)	Everything an Admin can do, plus receives escalation SMS for unacknowledged reports in their group.
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Design note: Citizens and admins live in separate Cognito user pools, not separate roles in one pool. This keeps sign-up policy different by design: citizens self-register, admin accounts are provisioned only by a system operator.

04 Functional Requirements

Each requirement is numbered for traceability against test cases.

4.1 Incident Reporting

ID	REQUIREMENT
FR-1.1	The system shall allow an authenticated citizen to submit a report containing: incident type, free-text description, location, witnesses (if available) and at least one photo.
FR-1.2	The system shall allow a citizen to submit a report anonymously, omitting identity from any data visible to admins.

FR-1.3	The system shall issue a presigned URL for direct photo upload to S3, rather than proxying the file through the API.
FR-1.4	When a report is queued due to lack of connectivity, the client shall immediately display the emergency hotline number associated with the selected incident type, so the citizen is not left relying solely on the queued submission for an urgent incident.
FR-1.5	The system shall return a tracking reference the reporter can use to check status without re-authenticating, for anonymous submissions.

4.2 Authentication & Access Separation

ID	REQUIREMENT
FR-2.1	The system shall authenticate citizens and admins via separate Cognito user pools.
FR-2.2	The system shall reject any admin-scoped API call carrying a citizen-pool token, and vice versa.
FR-2.3	The system shall assign each admin to exactly one routing group at account provisioning.

4.3 Notification & Dashboard

ID	REQUIREMENT
FR-3.1	The system shall notify every member of the routed admin group within seconds of report submission, via an in-app alert and an SES email.
FR-3.2	The in-app alert shall play an audible signal continuously for up to three minutes or until an admin acknowledges the report, whichever comes first.
FR-3.3	The admin dashboard shall list reports filtered to the admin's routing group, sorted by submission time.
FR-3.4	The dashboard shall visually cluster reports that share an incident type, location, and a submission window under ten minutes.

05 Routing & Escalation Logic

This is the core behavior distinguishing C.A.R.E from a generic ticketing form: each incident type reaches a specific authority, automatically, with a fallback if no one responds.

5.1 Type-to-group routing

Every incident type maps to exactly one primary routing group at submission time. The mapping is stored as configuration data, not hardcoded, so new incident types can be added without a deployment.

PROVISIONAL ROUTING TABLE SUBJECT TO CHANGE	
INCIDENT TYPE	ROUTING GROUP
Child abuse, Underage marriage	child-protection + police-general
Kidnapping, murder	police-major-crimes
Femicide, Gender-based violence, Domestic violence	gbv-response-unit
Burglary, Theft	police-general
Road accident	police-general + medical-emergency

5.2 Escalation ladder

ID	REQUIREMENT
FR-4.1	If no admin in the routed group acknowledges a report within 3 minutes, the system shall re-send the in-app and email alert to the same group.

FR-4.2	If the report remains unacknowledged 5 minutes after submission, the system shall send an SMS to the group's designated supervisor via the third-party escalation service.
FR-4.3	The escalation-service API credential shall be retrieved from Secrets Manager at invocation time and never stored in application code, environment variables in plaintext, or logs.
FR-4.4	Escalation shall only page admins currently marked on-duty; if no one in a group is on-duty, escalation shall proceed directly to the supervisor contact.

Scope boundary: C.A.R.E guarantees notification, not response. Placing an actual emergency call and dispatching a unit remains a human decision made outside the system, by design, this keeps accountability for real-world action with the authority, not the software.

06 Incident Status Lifecycle

ID	REQUIREMENT
FR-5.1	Reaching ACKNOWLEDGED shall stop any pending escalation for that report.

FR-5.2	Transitioning to RESOLVED shall require a non-empty case note from the acting admin.
FR-5.3	Every status transition shall be logged with the acting admin's ID and a timestamp, immutable after write.
FR-5.4	The reporter shall be able to view current status via their tracking reference, without seeing internal case notes.

07 External Interfaces

7.1 REST API (subjected to change)

METHOD	PATH	CALLER	PURPOSE
POST	/reports	Citizen	Create a report, receive a presigned S3 upload URL
GET	/reports/{id}	Citizen / Admin	Read one report (scoped by ownership or group)

GET	/reports?status=	Admin	List reports routed to caller's group
PATCH	/reports/{id}/acknowledge	Admin	Stop escalation, mark seen
PATCH	/reports/{id}/status	Admin	Advance lifecycle state, attach note
WSS	/alerts	Admin	Persistent connection for real-time in-app alerts

7.2 Third-party interfaces

One outbound integration: an SMS/push escalation provider, invoked only by the escalation Lambda, authenticated with a credential held in Secrets Manager and rotated independently of application deployments.

08 Non-Functional Requirements

CATEGORY	REQUIREMENT

Performance	An admin in the routed group receives the in-app alert within 5 seconds of submission under normal load.
Availability	Notification delivery channels (in-app, email, SMS) shall fail independently. A breakdown in one shall not block the others.
Security	No credential, API key, or secret shall appear in source control, logs, or environment variables in plaintext.
Privacy	Child-abuse and GBV report content shall be restricted to the routing group assigned to that type, not all admins.
Usability	The citizen-facing form shall be available in French and English, switchable without losing in-progress input.
Resilience	Report submission shall succeed from a client with intermittent connectivity via local queuing (FR-1.4).
Auditability	Every status change and escalation event is retained as an immutable log entry (FR-5.3).

09 Architecture

9.1 System overview

[still to include an architecture after implementing other design diagrams]

9.2 Submission-to-escalation sequence

[still figuring it out perfectly]

9.3 Infrastructure as Code

The system is provisioned with AWS CDK, split into independent stacks: AuthStack, DataStack, ApiStack, AlertingStack, so each concern can be deployed and reasoned about separately.

10 Appendix

10.1 Provisional incident type list

Pending field data on incident frequency in Cameroon. Current working list: Kidnapping, Burglary, Theft, Child abuse, Underage marriage, Road accident, Murder, Femicide, Gender-based violence. Stored as configuration, not code, so this list may grow without a redeployment.

10.2 Explicitly out of scope (v1)

- Integration with existing government case-management systems
- Geographic/jurisdictional routing (routing is by incident type only in this version) with the assumption that every community in Cameroon make use of same emergency contact if face with similar incident.
- Automated placement of emergency phone calls

